Physics-Aligned Gating for Mixture-of-Experts Forecasting under Regime Shifts: Case Studies in Wind Farms and ERA5

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Abstract

Forecast errors in non-stationary spatio-temporal physical systems tend to accumulate near regime transitions, where one observation stream may mix multiple local response laws. Dense graph-temporal forecasters reduce average error by exploiting transport and persistence, yet they still apply a shared mapping across those transitions. Mixture-of-experts routing is therefore attractive, but routing learned only from prediction loss can collapse when regime markers are weakly expressed or partly obscured by control logic. We test a physics-aligned gating framework built on a directed-diffusion GRU encoder and a node-level MoE gate regularized by load balancing, regime-anchor alignment, graph smoothness, and a boundary-focused forcing term. ERA5 reanalysis and the KDD Cup 2022 wind-farm SCADA benchmark are used as contrasting observability settings: atmospheric stability is strongly expressed in sensible heat flux in ERA5, whereas blade-pitch control partly obscures the aerodynamic boundary in WTB. In ERA5, MoE only gives the lowest headline forecasting error, while the corrected model mainly improves gate-regime agreement to NMI/ARI = 0.5282/0.7147 with test RMSE 0.8679. In WTB, the strongest corrected routing configuration is not the deepest loss stack but the boundary-focused variant that combines balancing, regime alignment, and explicit forcing; that configuration reaches overall RMSE 243.97, switch-window RMSE 248.84, pitch-control RMSE 305.10, and gate-regime NMI/ARI 0.8511/0.9093. Across the two settings, ERA5 acts as an observability contrast for routing semantics, whereas WTB provides the stronger evidence that control-confounded systems need explicit physical correction at the gate.

Keywords: Mixture of experts; regime shift; physics-aligned learning; wind-power forecasting; ERA5; spatio-temporal graph learning.

1 Introduction

Forecasting errors in renewable-rich systems are not uniformly costly. The most consequential errors often occur in short transition windows, when the local response law changes faster than operators can retune dispatch, reserves, curtailment, or turbine control. In wind farms, this difficulty appears near the MPPT-to-pitch boundary and during wake-affected inflow shifts, where power no longer follows wind speed through one smooth mapping [Khodayar and Wang, 2019, Zhou et al., 2024, Haq et al., 2025]. A similar issue appears in the lower atmosphere, where stable-to-convective transitions reorganize transport and surface forcing over short horizons that matter for subsequent prediction [Hersbach et al., 2020]. The analysis therefore focuses on that transition-window difficulty.

Spatio-temporal graph forecasting has already improved average predictive accuracy in many correlated systems. DCRNN, STGCN, Graph WaveNet, ASTGCN, AGCRN, and MTGNN are representative examples [Li et al., 2018, Yu et al., 2018, Wu et al., 2019, 2020, Guo et al., 2019, Bai et al., 2020]. Together, they show that graph propagation and temporal encoding can model asymmetric dependence and dynamic correlation better than dense sequence baselines. The same idea has been carried into wind forecasting, where graph structure helps represent wake interaction, transport, and turbine coupling [Khodayar and Wang, 2019, Park and Park, 2019, Yu et al., 2020, Daenens et al., 2025]. Yet benchmarking in this literature is still dominated by overall MAE, RMSE, or MAPE, rather than by the short windows where the physical mechanism changes [Li et al., 2018, Wu et al., 2019, Khodayar and Wang, 2019, Daenens et al., 2025]. These studies show that graph-temporal structure helps, but shared mapping remains the default.

Mixture-of-experts offers a different answer to heterogeneity. Classical and modern MoE models use

routing to allocate different inputs to different predictors, which can increase effective capacity while keeping computation sparse [Jacobs et al., 1991, Jordan and Jacobs, 1994, Shazeer et al., 2017, Fedus et al., 2022]. Recent sequence-forecasting work has started to import the same specialization idea into large time-series models [Shi et al., 2025]. In parallel, physics-guided learning has shown that prior knowledge can enter the pipeline through synthetic pretraining or feature construction. It can also enter through architecture design or physics-based losses and constraints [Karpatne et al., 2017, Read et al., 2019, Park and Park, 2019, Raissi et al., 2019, Karniadakis et al., 2021, Parsa, 2025]. Together, these lines of work show that specialization and physical prior can each help under heterogeneity, but they rarely meet at the routing layer itself.

Three gaps remain for the problem studied here. First, regime-transition error is rarely isolated as a first-class target, even though average metrics can hide failure near operating boundaries [Khodayar and Wang, 2019, Wu et al., 2020, Daenens et al., 2025]. Second, most physics-guided models regularize latent states or outputs, not the gate that decides which local predictor becomes active [Raissi et al., 2019, Karniadakis et al., 2021, Zehtabiyan-Rezaie et al., 2023]. Third, in control-confounded observations such as wind-farm SCADA, prediction loss alone can still produce a low-error gate that is mechanically uninformative. The unresolved issue is therefore not whether routing can reduce average error, but whether routing can recover a meaningful operating partition near mechanism boundaries.

This study proposes a physics-aligned gating framework for non-stationary spatio-temporal forecasting. The model combines a directed-diffusion GRU encoder with a node-level MoE gate and a small set of routing regularizers that target expert starvation, weak regime alignment, local inconsistency, and boundary ambiguity. WTB and ERA5 are used as deliberately contrasted observability settings: the operating boundary is partly obscured by control action in WTB, whereas the thermodynamic marker is directly expressed in ERA5. The study makes three specific contributions. It recasts non-stationary graph forecasting as a routing problem around mechanism transitions. It also moves physics guidance from the output side to the gate itself and uses the WTB-ERA5 contrast to show how the needed correction strength depends on observability.

2 Related Work

2.1 Spatio-temporal graph forecasting

Most spatio-temporal graph forecasting studies arrive at the same empirical finding: once spatial dependence is encoded, average forecast error drops. DCRNN, STGCN, and Graph WaveNet established this pattern in correlated forecasting problems by combining graph propagation with recurrent or temporal-convolutional encoders [Li et al., 2018, Yu et al., 2018, Wu et al., 2019]. Later models such as ASTGCN, AGCRN, and MTGNN relaxed the assumption of a fixed graph and showed that adaptive dependence learning can further improve average accuracy [Guo et al., 2019, Bai et al., 2020, Wu et al., 2020]. The same logic has been used in wind forecasting, where graph construction helps represent wake interaction, transport, and turbine coupling [Park and Park, 2019, Yu et al., 2020, Daenens et al., 2025]. These studies establish that graph structure is effective for average prediction, but they still treat a shared mapping as the default across mechanism transitions.

2.2 Mixture-of-experts and routing for sequence modeling

The MoE literature establishes a second empirical point: heterogeneous inputs can often be modeled better by specialized predictors than by one dense mapping. Early adaptive and hierarchical MoE models showed how routing can partition the input space into regions with different local experts [Jacobs et al., 1991, Jordan and Jacobs, 1994]. Later sparse-routing work showed how this idea scales, while also exposing router collapse, expert starvation, and the need for explicit balancing [Shazeer et al., 2017, Fedus et al., 2022]. Recent sequence forecasting work has started to use the same specialization principle in large time-series models [Shi et al., 2025]. The missing step for our setting is not sparse routing itself,

Table 1. Core notation used in the routing and loss definitions.

Symbol	Meaning	Dimension
$\mathbf{x}_{i,t}$	Node feature vector at node i, time t	$\mathbb{R}^F$
$\mathbf{X}_{t-H+1:t}$	Input history window	$\mathbb{R}^{H\times N\times F}$
$y_{i,t}$	Scalar target	$\mathbb{R}$
$\mathbf{h}_{i,t}$	Encoded node context	$\mathbb{R}^d$
$\mathbf{a}_{i,t}$	Physics anchor vector used by the gate	$\mathbb{R}^q$
$\mathbf{z}_{i,t}$	Gate logits	$\mathbb{R}^E$
$\mathbf{g}_{i,t}$	Gate probabilities after softmax	$\mathbb{R}^E$
$f_e(\cdot)$	Expert-e prediction head	$\mathbb{R}^d \to \mathbb{R}^P$
$R_{i,t}$	Primary regime anchor label	discrete
$W_{i,t}$	Auxiliary wake flag in WTB	$\{0, 1\}$
$\mathscr{A}_t(i,j)$	Edge weight from node j to node i at time t	scalar
E	Number of experts	scalar
K	Top-K used in the balancing loss	scalar

but a router that is constrained by physically meaningful mechanism boundaries rather than by prediction loss alone.

2.3 Physics-informed and physics-guided learning in spatio-temporal systems

Physics-guided learning has established a third point: prior knowledge can be injected at several stages of the pipeline. Theory-guided data science and process-guided deep learning emphasize synthetic data, feature construction, pretraining, and post-hoc constraints [Karpatne et al., 2017, Read et al., 2019]. PINN-style methods and later reviews show the same idea at the optimization, architecture, and loss levels [Raissi et al., 2019, Karniadakis et al., 2021, Parsa, 2025]. In renewable and atmospheric applications, physics-induced GNNs, lidar-assisted wind-field prediction, frequency-domain PINNs, and physics-guided wind-farm power models have improved robustness or interpretability by injecting physical information into the model pipeline [Park and Park, 2019, Zhang and Zhao, 2021, Li et al., 2025, Zehtabiyan-Rezaie et al., 2023]. What remains underexplored is whether the same prior should directly constrain the routing decision itself.

3 Methodology

3.1 Problem setup and notation

The forecasting task is written to separate two decisions that dense models often merge: predicting the future and deciding which local mapping should be active at the anchor time. Let $G = (V, E)$ denote a spatial graph with $N = |V|$ nodes. For each node $i \in V$ and time step t, we observe a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predict a scalar target $y_{i,t} \in \mathbb{R}$. Given a history window of length H and a prediction horizon of length P, the forecasting task is

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathscr{F}\left(\mathbf{X}_{t-H+1:t}, \mathscr{A}_{t-H+1:t}\right),$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H\times N\times F}$ and $\mathscr{A}_{t-H+1:t}$ denotes either a time-varying directed graph sequence (WTB) or a static graph repeated over time (ERA5). Invalid or missing targets are excluded by a supervision mask.

Table 1 summarizes the notation used in the routing and loss definitions. Routing is node-level rather than graph-level: each node at each anchor time receives its own gate distribution $\mathbf{g}_{i,t}$. This is essential because nearby turbines or grid cells can move into different local regimes within the same sequence.

3.2 Shared architecture

3.2.1 Physical graph construction

The graph is treated as part of the physical problem specification. In WTB, the graph should expose likely wake pathways rather than simple proximity. In ERA5, the graph only needs to preserve geographically local dependence because the thermodynamic regime marker is already visible in the state. Exact graph constants are listed in the Appendix tables.

For WTB, each turbine has a fixed coordinate $\mathbf{p}_i = (x_i, y_i)$. A candidate set is first built by retaining the k_{nn} nearest turbines within a maximum radius $d_{\max}$. Let $\Delta\mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$ and let $\theta_{i,t}$ denote the local meteorological wind-from direction. The downstream unit vector is

$$\mathbf{u}_{i,t} = \begin{bmatrix} \sin(\theta_{i,t} + \pi) \\ \cos(\theta_{i,t} + \pi) \end{bmatrix}.$$

which gives the streamwise and cross-stream distances

$$d^{\parallel}_{ij,t} = \Delta\mathbf{p}^{\top}_{ij}\mathbf{u}_{i,t}, \qquad d^{\perp}_{ij,t} = \left|\Delta p^{x}_{ij} u^{y}_{i,t} - \Delta p^{y}_{ij} u^{x}_{i,t}\right|.$$

An edge is activated only when turbine j falls inside a downstream cone,

$$\mathbb{I}^{\mathrm{cone}}_{ij,t} = \mathbf{1}\left[d^{\parallel}_{ij,t} > 0 \;\wedge\; \arctan\left(\frac{d^{\perp}_{ij,t}}{\max(d^{\parallel}_{ij,t}, 10^{-6})}\right) \le \phi\right],$$

and the resulting wake weight is

$$\tilde{\mathscr{A}}_t(i,j) = \exp\left(-\frac{d^{\parallel}_{ij,t}}{\alpha}\right)\exp\left(-\frac{|d^{\perp}_{ij,t}|}{\beta}\right)\mathbb{I}^{\mathrm{cone}}_{ij,t},$$

If wind direction is missing, the implementation falls back to a static proximity weight

$$\mathscr{A}^{\mathrm{static}}(i,j) = \exp\left(-\frac{\|\Delta\mathbf{p}_{ij}\|_2}{d_{\max}}\right).$$

Only the strongest M inbound weights are retained for each target and time step. The same weights define a wake score,

$$s^{\mathrm{wake}}_{i,t} = \sum_j \mathscr{A}_t(i,j),$$

which is later used as an auxiliary routing indicator.

ERA5 uses a simpler graph because the main uncertainty is not hidden directionality but local transport over the retained patch. Each grid point is treated as a node with geographic coordinate (φ_i, λ_i), and the great-circle distance is computed by the Haversine formula

$$d_{ij} = 2R\arctan\left(\frac{\sqrt{a_{ij}}}{\sqrt{1 - a_{ij}}}\right),$$

where $R = 6371$ km and

$$a_{ij} = \sin^2\left(\frac{\varphi_j - \varphi_i}{2}\right) + \cos(\varphi_i)\cos(\varphi_j)\sin^2\left(\frac{\lambda_j - \lambda_i}{2}\right).$$

We then retain a symmetric k_{nn}-nearest-neighbor graph and define

$$\mathcal{A}(i,j) = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right)\mathbf{1}[j \in \mathcal{N}_{k_{\mathrm{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\mathrm{nn}}}(j)],$$

where σ is the median retained neighbor distance on the training graph. The design is intentionally simpler than WTB because the ERA5 regime marker is already visible in the observed state.

3.2.2 Directed-diffusion GRU encoder and node-level gate

The encoder is kept modest so that any change in performance can be attributed to routing and regularization rather than to a stronger backbone. Each input window is concatenated with its feature-missingness mask before projection. Two directed diffusion blocks then aggregate self, inbound, and outbound messages on each graph snapshot. Let $\mathbf{X}_t^{(\ell)}$ denote the node state at layer ℓ and time t. A diffusion block updates each node by

$$\mathbf{X}_t^{(\ell+1)} = \mathrm{LN}\Big(\mathbf{W}_{\mathrm{self}}^{(\ell)}\mathbf{X}_t^{(\ell)} + \mathbf{W}_{\mathrm{in}}^{(\ell)}\mathrm{Agg}_{\mathrm{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{W}_{\mathrm{out}}^{(\ell)}\mathrm{Agg}_{\mathrm{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)}\mathbf{X}_t^{(\ell)}\Big),$$

where $\mathrm{Agg}_{\mathrm{in}}$ and $\mathrm{Agg}_{\mathrm{out}}$ are weighted sums over inbound and outbound neighbors. The spatially updated sequence is then processed by a GRU to produce a node-wise context vector $\mathbf{h}_{i,t}$.

Each expert head f_e maps $\mathbf{h}_{i,t}$ to a P-step forecast. Routing remains dense during the forward pass. Every expert contributes, but its contribution is weighted by the soft gate distribution. The gate receives both the learned context and a small physics anchor,

$$\mathbf{z}_{i,t} = \mathrm{MLP}_{\mathrm{gate}}\big([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}]\big), \qquad \mathbf{g}_{i,t} = \mathrm{softmax}\left(\frac{\mathbf{z}_{i,t}}{\tau}\right),$$

and the forecast is

$$\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^{E} g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t}).$$

WTB uses the anchor $\mathbf{a}_{i,t} = [\texttt{Wspd}, \texttt{Pab_mean}, s^{\mathrm{wake}}, \texttt{Patv}]$, while ERA5 uses $\mathbf{a}_{i,t} = [\texttt{sshf}, \texttt{t2m}, \texttt{wind_speed}, \Delta\texttt{sshf}$ The difference reflects the observability contrast: WTB needs help to recover a partly hidden operating boundary, whereas ERA5 already exposes the relevant thermodynamic marker in the state.

3.3 Physics-aligned routing regularizers

Prediction loss alone does not identify the gate in a unique way when several experts can fit averaged dynamics similarly well [Shazeer et al., 2017, Fedus et al., 2022]. The problem is especially acute in WTB, where inflow variation, wake disturbance, and pitch control are observed together. We therefore use a small set of routing regularizers, each aimed at a different weakness of prediction-only routing. The total objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{pred}} + \lambda_{\mathrm{bal}}\mathcal{L}_{\mathrm{bal}} + \lambda_{\mathrm{align}}\mathcal{L}_{\mathrm{align}} + \lambda_{\mathrm{force}}\mathcal{L}_{\mathrm{force}} + \lambda_{\mathrm{aux}}\mathcal{L}_{\mathrm{aux}} + \lambda_{\mathrm{smooth}}\mathcal{L}_{\mathrm{smooth}},$$

with inactive terms set to zero in settings where they are not used. Exact weights are moved to the Appendix. The important methodological point is that WTB needs orders-of-magnitude stronger alignment and forcing than ERA5 because the operating boundary is weakly observable in the raw state.

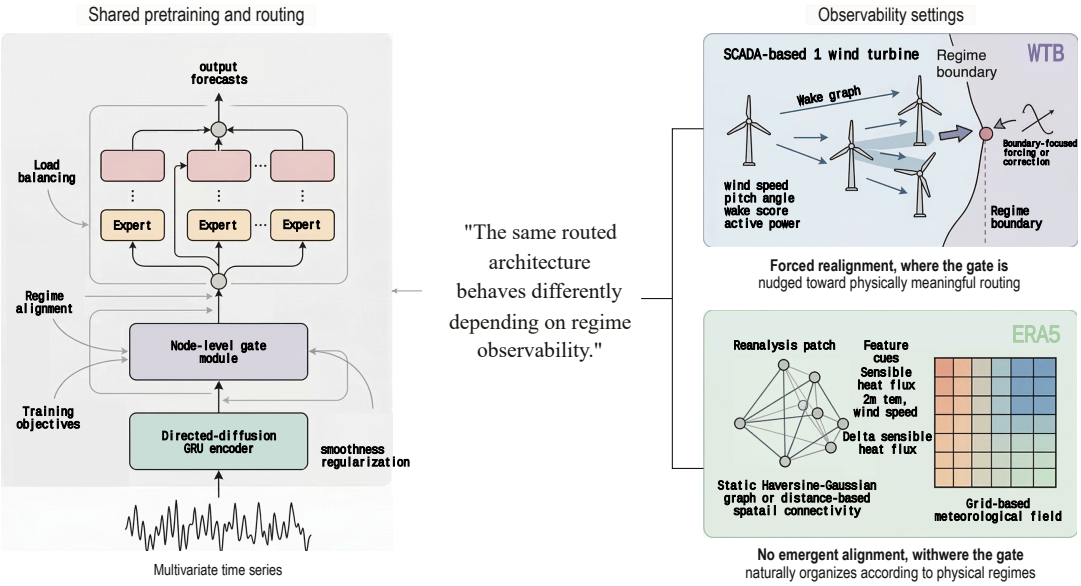

Figure 1: Physics-aligned regime-aware MoE. The two datasets share the same directed-diffusion GRU encoder and node-level MoE routing mechanism. WTB uses a dynamic wake graph and a boundary-focused forcing term, whereas ERA5 uses a Haversine-Gaussian graph and a thermodynamic regime anchor.

3.3.1 Prediction objective

The prediction loss remains the anchor of the optimization, because any routing correction becomes self-defeating if it no longer serves the forecast itself. We therefore retain the masked mean-squared-error loss

$$\mathscr{L}_{\text{pred}} = \frac{\sum_{i,t,\tau} m^{y}_{i,t+\tau} \left(\hat{y}_{i,t+\tau} - y_{i,t+\tau}\right)^2}{\sum_{i,t,\tau} m^{y}_{i,t+\tau}},$$

where $m^{y}_{i,t+\tau}$ is the target-validity mask. All routing regularizers remain subordinate to this prediction objective.

3.3.2 Load balancing

A slightly favored expert can dominate early training and starve the others before regime-specific structure has time to emerge. A balancing term counteracts that early collapse. Let $\mathbf{g}_n$ denote the gate distribution for the n-th valid routed sample in a mini-batch of size B. The soft importance of expert e is

$$I_e = \frac{1}{B}\sum_{n=1}^{B} g_n^{(e)}, \qquad P_e = \frac{I_e}{\sum_{r=1}^{E} I_r}.$$

The empirical top-K load is

$$f_e = \frac{1}{B}\sum_{n=1}^{B} \mathbf{1}[e \in \text{TopK}(\mathbf{g}_n)].$$

The balancing loss is

$$\mathscr{L}_{\text{bal}} = E \sum_{e=1}^{E} f_e P_e - 1.$$

This term does not decide which expert should own which regime. It only prevents early expert starvation.

3.3.3 Regime-anchor alignment

Balanced expert usage can still produce a physically meaningless partition. The next term aligns the gate with the coarsest regime structure that can be identified with high confidence. Let $\mathbf{z}_{i,t}^{(1:C)}$ denote the first C gate logits, and let $R_{i,t}$ and $M_{i,t}$ be the primary regime label and its validity mask. The alignment loss is

$$\mathscr{L}_{\text{align}} = \frac{1}{|\Omega|} \sum_{(i,t)\in\Omega} \text{CE}\left(\mathbf{z}_{i,t}^{(1:C)}, R_{i,t}\right), \qquad \Omega = \{(i,t) : M_{i,t} = 1\},$$

with inverse-frequency class weights estimated on the training split. Alignment does not fully supervise the router. It only stabilizes the coarse physical partition where the labels are reliable.

3.3.4 Boundary-focused forcing (WTB only)

The MPPT-to-pitch boundary in WTB remains ambiguous even after coarse regime alignment because control action partly masks the mechanical transition. A focused forcing term therefore acts only on the most informative MPPT and pitch-control samples. Let

$$Y_{i,t}^{\text{force}} = \begin{cases} 0, & R_{i,t}^{\text{wtb}} = 1, \\ 1, & R_{i,t}^{\text{wtb}} = 2. \end{cases}$$

Using the MPPT-aligned and pitch-aligned expert logits, we define

$$\mathscr{L}_{\text{force}} = \frac{1}{|\Omega_{\text{force}}|} \sum_{(i,t)\in\Omega_{\text{force}}} \text{CE}\left(\begin{bmatrix} z_{i,t}^{(\text{mppt})} \\ z_{i,t}^{(\text{pitch})} \end{bmatrix}, Y_{i,t}^{\text{force}} \right),$$

where $\Omega_{\text{force}} = \{(i,t) : R_{i,t}^{\text{wtb}} \in \{1,2\}\}$. This term repairs the mechanically important boundary instead of supervising the full gate everywhere.

3.3.5 Wake auxiliary supervision and graph smoothness

After the MPPT-to-pitch boundary is repaired, wake-sensitive samples can still be mixed with non-wake samples inside the same operating regime. A wake auxiliary label identifies that residual ambiguity in WTB, while graph smoothness discourages noisy neighbor-to-neighbor gate jumps in both settings. The wake auxiliary loss is

$$\mathscr{L}_{\text{aux}} = \frac{1}{|\Omega_{\text{wake}}|} \sum_{(i,t)\in\Omega_{\text{wake}}} \text{BCE}\left(z_{i,t}^{(\text{wake})}, W_{i,t}\right),$$

where Ω_{wake} contains only MPPT and pitch-control samples for which the wake flag is defined. The smoothness term is

$$\mathscr{L}_{\text{smooth}} = \frac{\sum_{i,j} \mathscr{A}_t(i,j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{i,j} \mathscr{A}_t(i,j)}.$$

The auxiliary term is a WTB-specific repair for wake mixing, while the smoothness term encourages locally coherent routing without prescribing a global partition.

3.4 Dataset-specific anchors and labels

The gate is not supervised everywhere. Instead, it is anchored only where the physical interpretation is clearest, and the ambiguous remainder is organized by prediction loss plus the routing regularizers.

3.4.1 WTB operating regimes and wake labels

In WTB, the primary operating boundary is the transition from MPPT to pitch control. Let

$$\bar{p}_{i,t} = \frac{1}{3}\left(p_{i,t}^{(1)} + p_{i,t}^{(2)} + p_{i,t}^{(3)}\right).$$

Using a cut-in threshold u_{idle}, a pitch-transition threshold u_{rated}, and a pitch-angle threshold p_{th}, we define

$$R_{i,t}^{\text{wtb}} = \begin{cases} 0, & \text{if } \texttt{Wspd}_{i,t} < u_{\text{idle}} \quad \text{(idle)}, \\ 1, & \text{if } u_{\text{idle}} \le \texttt{Wspd}_{i,t} \le u_{\text{rated}} \ \wedge \ \bar{p}_{i,t} < p_{\text{th}} \quad \text{(MPPT)}, \\ 2, & \text{if } \texttt{Wspd}_{i,t} > u_{\text{rated}} \ \wedge \ \bar{p}_{i,t} \ge p_{\text{th}} \quad \text{(pitch-control)}, \\ 3, & \text{otherwise} \quad \text{(transition)}. \end{cases}$$

Only the first three classes are used in direct alignment. Transition samples are retained for analysis but masked out of label supervision through

$$M_{i,t}^{\text{wtb}} = \mathbf{1}[R_{i,t}^{\text{wtb}} \neq 3].$$

Wake interference is handled separately because it is spatial rather than mechanical. Let $q_{0.75}^{\text{wake}}$ denote the upper-quartile threshold of the wake score over valid operating states in the training split. Then

$$W_{i,t} = \begin{cases} 1, & \text{if } s_{i,t}^{\text{wake}} \ge q_{0.75}^{\text{wake}} \ \wedge \ R_{i,t}^{\text{wtb}} \in \{1,2\}, \\ 0, & \text{if } s_{i,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \ \wedge \ R_{i,t}^{\text{wtb}} \in \{1,2\}, \\ \emptyset, & \text{otherwise}. \end{cases}$$

The exact thresholds are reported in the Appendix tables.

3.4.2 ERA5 thermodynamic regimes

In ERA5, the anchor is built around the stable-to-convective transition. Surface sensible heat flux changes sign across that transition, and large flux gradients mark disturbed periods. Let $\Delta\texttt{sshf}_{i,t} = \texttt{sshf}_{i,t} - \texttt{sshf}_{i,t-1}$. We use a small-margin threshold $\varepsilon_{\text{sshf}}$ for $|\texttt{sshf}|$. We also use a high-quantile threshold $q_{0.95}^{\Delta}$ for $|\Delta\texttt{sshf}|$. The regime label is then defined by

$$R_{i,t}^{\text{era5}} = \begin{cases} 0, & \text{if } \texttt{sshf}_{i,t} > \varepsilon_{\text{sshf}} \ \wedge \ |\Delta\texttt{sshf}_{i,t}| \le q_{0.95}^{\Delta} \quad \text{(convective)}, \\ 1, & \text{if } \texttt{sshf}_{i,t} < -\varepsilon_{\text{sshf}} \ \wedge \ |\Delta\texttt{sshf}_{i,t}| \le q_{0.95}^{\Delta} \quad \text{(stable)}, \\ 2, & \text{otherwise} \quad \text{(transition/anomalous)}. \end{cases}$$

If sensible heat flux is unavailable, the same construction can fall back to an analogous rule based on the temporal gradient of 2 m temperature. The exact thresholds used in the reported experiments are again listed in the Appendix.

Table 2. In-family model comparison used for mechanism validation in the current encoder family.

Model	Routing	Extra terms	Params	Role
Dense (matched)	Single dense head	None	WTB 110,012 / ERA5 103,843	Shared-mapping reference
Unconstrained MoE	Node-level soft gate	Prediction loss only	WTB 110,012 / ERA5 103,843	Tests whether routed capacity alone is enough
Corrected routing model	Node-level soft gate	WTB: L_{bal} + L_{align} + L_{force}; ERA5: full corrected stack	WTB 110,012 / ERA5 103,843	Headline corrected routing comparator in each observability setting

4 Experimental Setup

4.1 Datasets and preprocessing

The experiment contrasts two observability settings rather than two interchangeable benchmarks. WTB is the control-confounded case. The KDD Cup 2022 benchmark contains 134 turbines and 245 days of 10 min SCADA measurements [Zhou et al., 2024]. However, the MPPT-to-pitch transition is only indirectly visible because inflow variation, wake disturbance, and blade-pitch control are mixed in the same stream.

The data are reorganized into synchronous tensors with $T = 35{,}280$ time steps, $N = 134$ turbines, and $F = 11$ dynamic inputs. Missing inputs are forward-filled within turbine and then completed with training-set means. Invalid or non-positive active power is retained only through its mask so that the model still sees operational irregularity without treating those points as supervised targets.

ERA5 is the signal-expressive case. We use three archived months over a fixed 16×16 hourly patch [Hersbach et al., 2020]. In this setting, the stable-to-convective transition is much easier to localize because sensible heat flux and its temporal variation are directly observed. The retained tensor has $T = 2208$ frames, $N = 256$ nodes, and eight input variables. Both datasets use the same history and horizon lengths, $H = 36$ and $P = 24$, so that differences in performance cannot be attributed to unequal context length. The chronological split is 180/30/35 days for WTB and 1325/441/442 frames for ERA5.

4.2 In-family comparison protocol

The comparison keeps the backbone family fixed and asks whether routing changes the forecast once shared-capacity explanations have been ruled out. It should therefore be read as mechanism validation within one encoder family rather than as an exhaustive benchmark sweep. The dense baseline uses the same directed-diffusion GRU encoder and replaces the expert mixture with one dense prediction head. The unconstrained MoE adds routed capacity without physical correction. The corrected-routing comparator is dataset-specific: the reported ERA5 comparator uses the full corrected stack, whereas the reported WTB comparator uses the boundary-focused variant selected by validation. Parameter budgets are matched within each dataset so that differences can still be attributed to routing and its regularization rather than to a stronger encoder.

For compactness, the tables below shorten the three main model names to **Dense (matched)**, **Unconstrained MoE**, and **Corrected routing**. The three displayed families are summarized in Table 2.

4.3 Training protocol and evaluation metrics

All models use the same training protocol. We use AdamW, early stopping on validation RMSE, gradient clipping, and mixed-precision training on a single CUDA-enabled GPU. Exact optimizer constants are reported in the Appendix so that the main text can stay focused on the comparison logic rather than on implementation detail. Dense and unconstrained MoE baselines are repeated across seeds where

reported. Some corrected configurations are currently available only as validated runs, so those entries are interpreted more as mechanism checks than as formal stability claims.

The metrics are chosen to match the scientific question. Overall MAE and RMSE test whether routing improves forecasting at all. Switch-window MAE and RMSE test whether the improvement survives where the local response law changes. Regime-wise RMSE shows where the error reduction concentrates. NMI, ARI, usage entropy, and confusion matrices then test whether the gate corresponds to physically interpretable structure rather than to an arbitrary latent split.

Figure 2 summarizes the physical setting before the forecasting results are introduced.

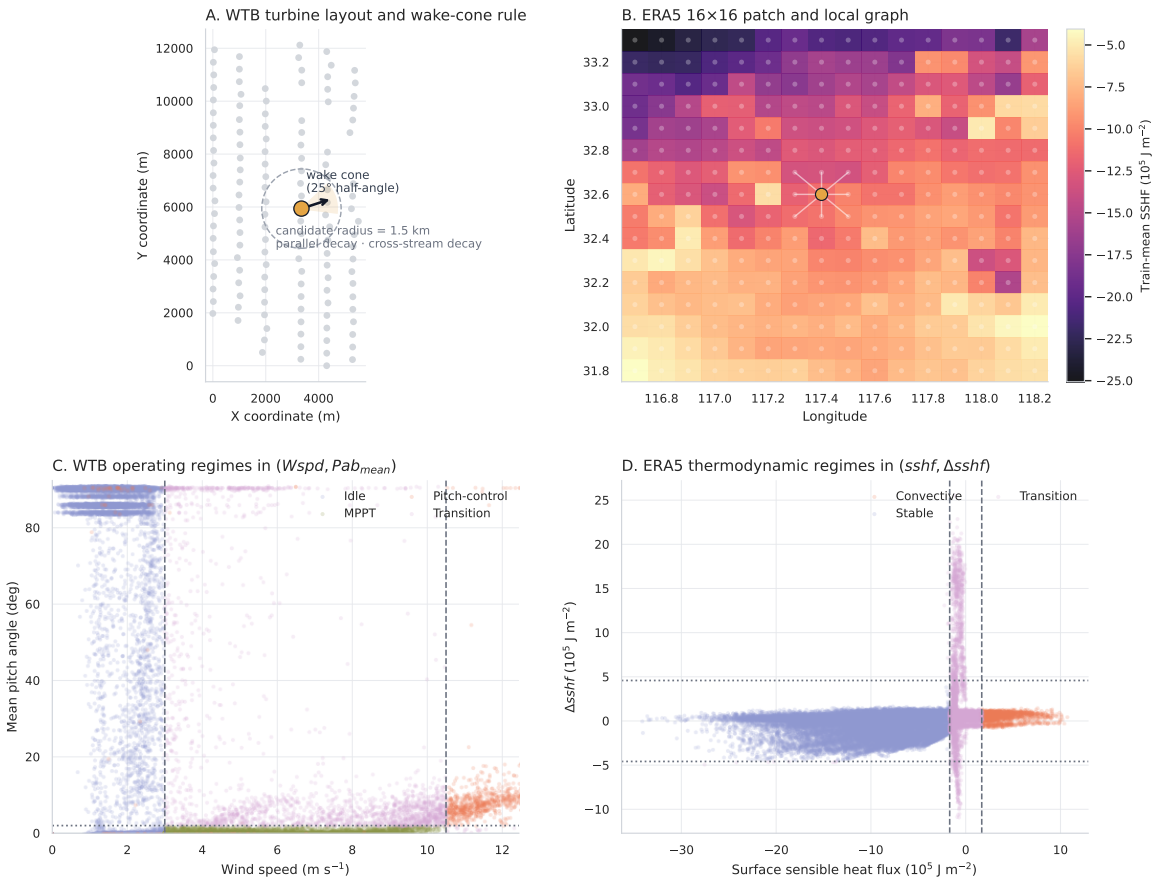

Figure 2: Dataset geometry and physical regime anchors. (A) WTB turbine layout with the schematic wake cone and the retained candidate radius used in the dynamic directed wake graph. (B) ERA5 16x16 patch with training-mean sensible heat flux and local Haversine-Gaussian graph connections around the central node. (C) WTB operating regimes in the $(Wspd, Pab_{mean})$ plane with fixed operating-rule boundaries. (D) ERA5 thermodynamic regimes in the $(sshf, \Delta sshf)$ plane with thresholds estimated from the training split.

5 Results

5.1 Overall performance

Table 3 and Figure 3 show that the two datasets support different claims. In WTB, corrected routing improves the headline forecast itself: overall RMSE drops from 258.47 for MoE only to 243.97, and switch-window RMSE drops from 260.82 to 248.84. In ERA5, the lowest headline error comes from MoE only, which reaches overall RMSE 0.8450 and switch-window RMSE 0.8158. The physics-aligned ERA5 model is slightly worse on those headline metrics, with 0.8679 and 0.8292, but it will matter

Table 3. Main forecasting performance within the current encoder family (mean ± std across seeds where repeated runs are available; single validated corrected runs are reported without uncertainty).

Model	Overall RMSE	Overall MAE	Switch RMSE	Switch MAE
Panel A. WTB				
Dense (matched)	261.16 ± 8.92	187.19 ± 11.09	264.41 ± 9.07	187.39 ± 11.34
MoE only	258.47 ± 9.65	182.82 ± 8.09	260.82 ± 12.13	182.01 ± 9.88
Corrected WTB routing	243.97	181.66	248.84	186.02
Panel B. ERA5				
Dense (matched)	0.9484	0.7482	0.9513	0.7501
MoE only	0.8450	0.6670	0.8158	0.6478
Physics-aligned MoE	0.8679	0.6836	0.8292	0.6587

Table 4. Regime-slice and transition-window performance (mean ± std across seeds where repeated runs are available; single validated corrected runs are reported without uncertainty).

Model	idle	MPPT	pitch-control	transition	switch-window
Panel A. WTB					
Dense (matched)	181.35 ± 18.27	272.97 ± 8.68	395.79 ± 43.05	343.23 ± 14.38	264.41 ± 9.07
MoE only	172.85 ± 14.42	272.85 ± 8.22	373.84 ± 69.71	340.19 ± 35.58	260.82 ± 12.13
Corrected WTB routing	172.37	260.22	305.10	305.04	248.84

Model	convective	stable	transition	switch-window
Panel B. ERA5				
Dense (matched)	0.5587	0.9622	0.9237	0.9513
MoE only	0.6647	0.8583	0.7774	0.8158
Physics-aligned MoE	0.5334	0.8885	0.7671	0.8292

later because its routing is much better aligned with the physical regimes. WTB is therefore the main boundary-repair result, whereas ERA5 functions as an observability contrast and positive-control case for routing semantics.

5.2 Transition-window and regime-slice performance

Table 4 shows why average error alone is not enough for this problem. In WTB, the key question is whether routing repairs the pitch-control boundary and the surrounding switch window. The corrected WTB routing variant reaches 305.10 pitch-control RMSE and 248.84 switch-window RMSE, while the unconstrained gate still mixes incompatible operating states and the deeper full stack over-regularizes the same boundary. In ERA5, the picture is narrower. Physics-aligned routing improves the convective slice and slightly improves the transition slice, but it does not beat MoE only on stable-state or switch-window headline error. This is consistent with mild calibration around a visible thermodynamic boundary rather than broad forecast repair.

5.3 Routing evidence

Table 5 shows that forecast improvement and routing recovery are not the same thing. This distinction is especially important for ERA5. Although MoE only has the best headline forecasting error there, its routing agreement is near zero, whereas the physics-aligned model raises NMI/ARI to 0.5282/0.7147. ERA5 therefore supports the interpretability side of the argument more than the headline-error side. In WTB, the corrected router reaches gate-regime NMI/ARI of 0.8511/0.9093 together with better forecast error, which is the stronger result. The WTB evidence indicates that explicit gate correction is needed to recover a physically meaningful partition in the control-confounded setting.

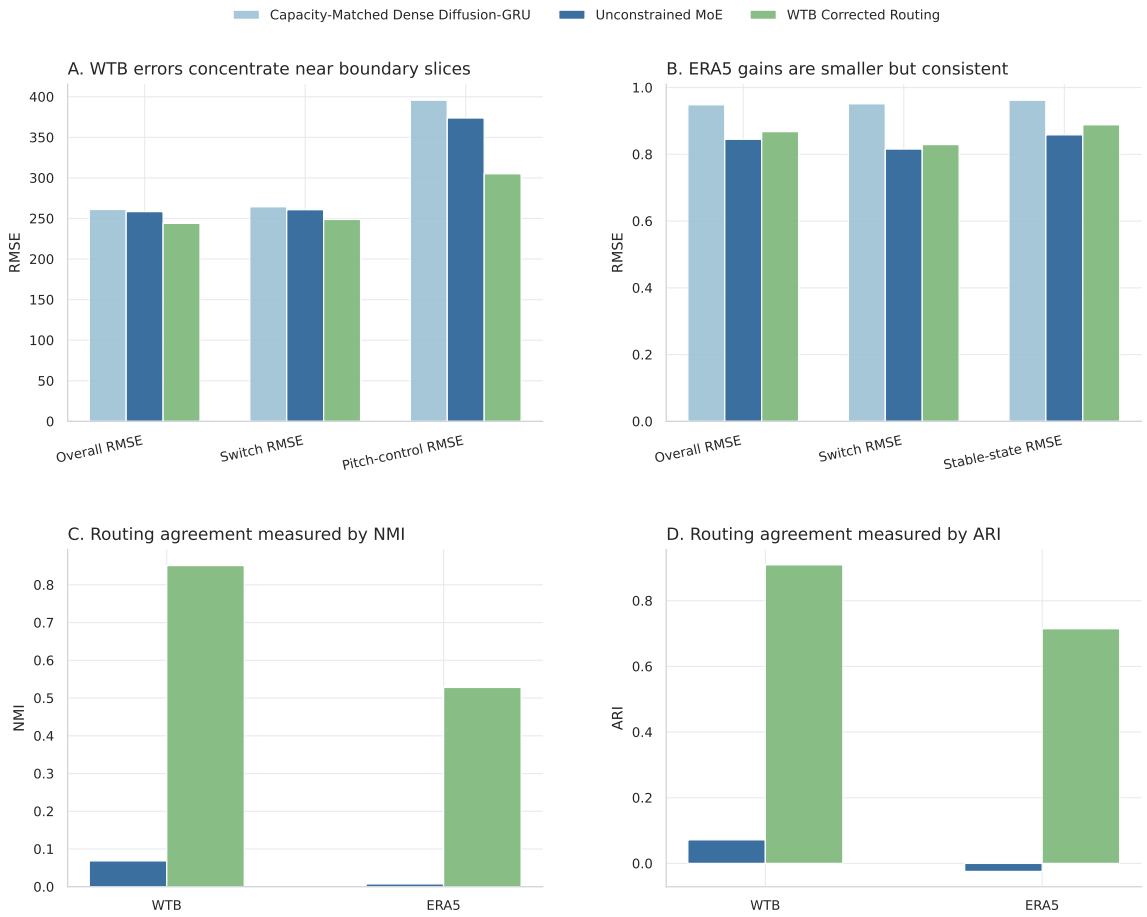

Figure 3: Summary of forecasting and routing outcomes. Panels A and B compress the main error comparison for WTB and ERA5. In WTB, the displayed corrected comparator is the boundary-forced routing variant rather than the deeper full stack. Panels C and D summarize routing agreement through NMI and ARI.

Table 5. Routing agreement and usage statistics (mean ± std across seeds where repeated runs are available; single validated corrected runs are reported without uncertainty).

Model	NMI	ARI	Entropy	Variance
Panel A. WTB				
MoE only	0.0685 ± 0.1370	0.0718 ± 0.1435	0.0000 ± 0.0000	0.1875 ± 0.0000
Corrected WTB routing	0.8511	0.9093	1.1120	0.0263
Panel B. ERA5				
MoE only	0.0075	-0.0244	0.8329	0.0619
Physics-aligned MoE	0.5282	0.7147	0.7990	0.0616

Table 6. WTB ablation on routing correction components (mean ± std across seeds).

Model	Overall RMSE	Switch RMSE	Pitch-control RMSE	NMI	
Dense (matched)	261.16 ± 8.92	264.41 ± 9.07	395.79 ± 43.05	–	
MoE only	258.47 ± 9.65	260.82 ± 12.13	373.84 ± 69.71	0.0685 ± 0.1370	0.0718
MoE + L_{bal}	–	–	–	–	
MoE + L_{align}	–	–	–	–	
MoE + L_{bal} + L_{align}	–	–	–	–	
MoE + L_{bal} + L_{align} + L_{force}	243.97	248.84	305.10	0.8511	
MoE + L_{bal} + L_{align} + L_{force} + L_{smooth}	–	–	–	–	
Physics-aligned MoE	265.18 ± 4.35	268.83 ± 4.15	404.78 ± 85.03	0.8658 ± 0.0001	0.9252

Figure 4 makes the WTB repair visible in the operating plane. After correction, the dominant expert regions align more cleanly with the MPPT and pitch-control structure, and the confusion matrices contract around the intended operating partition. The corrected gate is therefore not only more accurate; it is also more interpretable near the boundary that matters operationally.

Figure 5 shows the same contrast in time. In the WTB switch window, the corrected routing variant stays stable through the MPPT-to-pitch transition, while the dense and unconstrained models deviate more strongly. In ERA5, the expert weights evolve coherently with sensible heat flux and regime shading, which is what we would expect when the marker is already expressed in the state. These case studies connect the table-level metrics back to the physical mechanisms behind them.

5.4 Ablation

Table 6 should be read as a targeted decomposition around the completed WTB variants rather than as an exhaustive factorial ablation. Even with that limitation, the main pattern is still visible: the most important change comes from adding explicit forcing at the MPPT-to-pitch boundary. The later wake-plus-smoothness refinement does not improve the headline wind-farm result and instead exposes an over-regularized failure mode in the full WTB stack. The present ablation therefore supports a boundary-focused correction recipe as a practical repair. It does not yet provide a full marginal ranking of every regularizer in isolation.

6 Discussion

Across WTB and ERA5, the results indicate that correction strength should scale with observability, but not every dataset rewards correction in the same way. In ERA5, the regime marker is already visible in sensible heat flux and its temporal variation. There, MoE only already gives the best headline forecast error, while the corrected gate mainly improves regime alignment and interpretability. In WTB, the same transition is entangled with wake interaction and blade-pitch control, so the gate needs explicit boundary repair to recover a useful operating partition and to improve the forecast itself. The WTB-ERA5 contrast

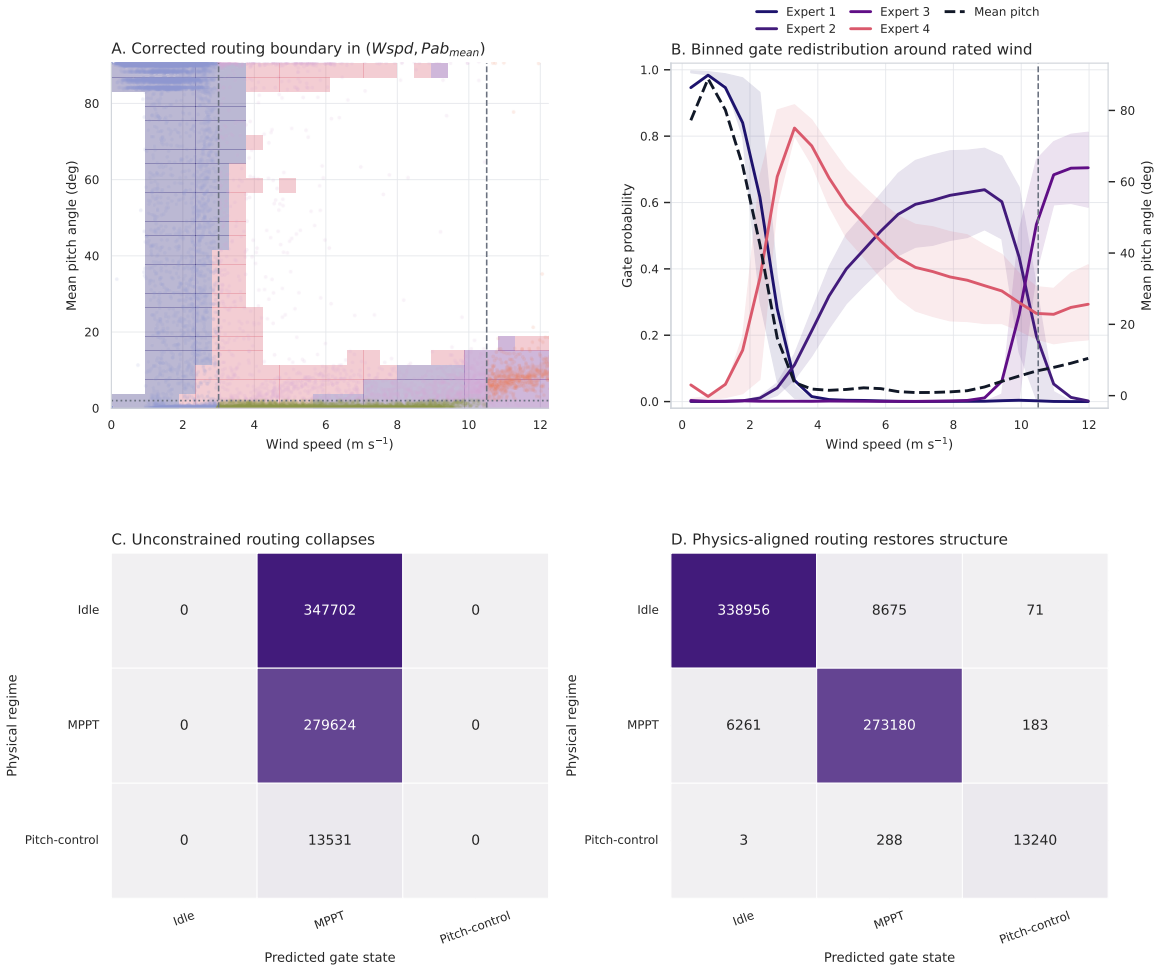

Figure 4: WTB routing diagnostics. Panel A maps the dominant expert in the $(Wspd, Pab_{mean})$ plane for the corrected WTB routing variant, with the sample cloud colored by physical operating regime. Panel B shows the binned gate redistribution against wind speed together with the mean pitch curve. Panels C and D compare the unconstrained and corrected confusion matrices.

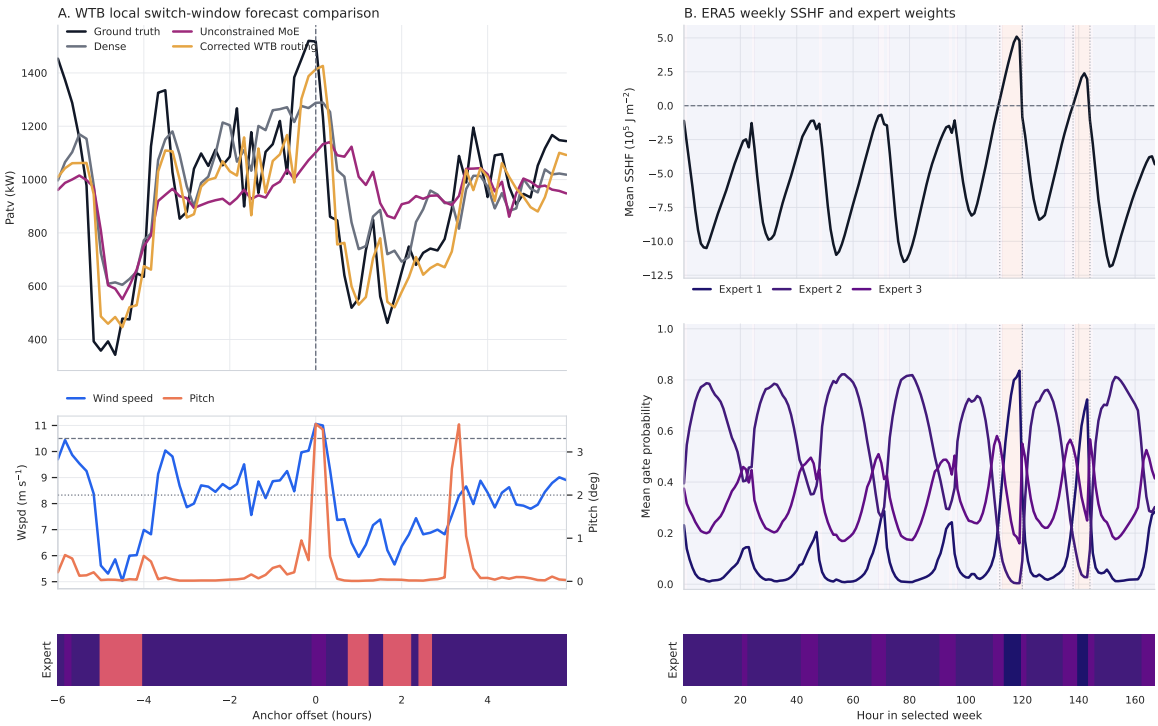

Figure 5: Dual case studies for correction and emergence. Panel A shows a local WTB switch window in which the corrected WTB routing variant remains stable while the dense and unconstrained models deviate more strongly near the MPPT-to-pitch transition; wind speed, pitch angle, and the dominant expert strip are plotted underneath. Panel B shows the ERA5 positive-control view, where sensible heat flux, regime shading, and expert weights evolve coherently over a selected week.

is therefore not evidence that one fixed loss stack is universally optimal. It is evidence that the gate should be corrected in proportion to how directly the regime boundary is expressed in the observations.

For operational forecasting, physics may need to constrain the routing decision itself. Many physics-guided models act on outputs or latent states, which is appropriate when the main task is to preserve physical consistency after prediction. Our setting is different. Here the central question is which local predictor should take responsibility near a mechanism change. In operational wind-farm practice, the MPPT-to-pitch boundary is often tuned conservatively because errors there are costly. The boundary-focused forcing term can be read as encoding that practice at the router rather than at the output head.

7 Limitations

The WTB correction still depends on threshold-based pseudo-labels derived from wind speed and mean pitch angle. Those thresholds are physically motivated and consistent with the implemented data pipeline, but they still impose a hand-crafted boundary definition. The method should therefore be read as boundary-aware regularized routing rather than as fully unsupervised regime discovery.

The comparison is intentionally bounded to the current encoder family. Capacity matching isolates the effect of routing within that family, but it is not a substitute for broader comparison against stronger dense graph-transformer, operator-style, or general time-series backbones. Extending the baseline set is therefore a next validation step, not a claim established here.

The WTB ablation is also partial rather than exhaustive. Several intermediate combinations in Table 6 were not run in the present study, so the ablation supports a practical boundary-forced recipe more strongly than a clean marginal decomposition of every regularizer.

Statistical reliability is still uneven across model families. Dense and unconstrained MoE baselines are reported with repeated seeds where available, but several corrected configurations remain single validated runs. The current evidence is therefore stronger on mechanism illustration than on initialization-robust significance claims for every corrected variant.

The paper also studies only two observability settings. WTB and ERA5 are sufficient for the present contrast between weakly and strongly expressed regime markers, but more industrial and geophysical case studies would be needed before generalizing the same interpretation beyond these two settings.

8 Conclusion

This paper studies non-stationary spatio-temporal forecasting as a routing problem around mechanism transitions. Within the current encoder family, the clearest forecast-side evidence comes from WTB, where failure concentrates near a control-confounded operating boundary and corrected routing repairs both error and partition quality. ERA5 plays a different role. It shows that when the regime marker is already visible, physics-aligned routing mainly improves the semantic quality of the gate rather than the headline forecasting score.

When the regime marker is visible, as in ERA5, routing regularization mainly calibrates an already available partition. When the marker is partly obscured by control logic, as in WTB, physics guidance at the gate acts as a repair for a weakly identified partition. In that setting, the most effective repair can come from a boundary-focused intermediate stack rather than from the deepest regularized model. The main message is therefore specific: in physical forecasting problems with multiple local response laws, physics should sometimes constrain not only what is predicted, but also which predictor becomes responsible.

Table A1. Shared architecture, graph, and training constants used in the reported experiments.

Item	WTB	ERA5	Role
History / horizon (H, P)	$(36, 24)$	$(36, 24)$	Input and forecast lengths
Gate softmax temperature τ	0.7	0.7	Soft routing sharpness
Top-K in L_{bal}	1	1	Balancing statistic only
Hidden size d	64	64	Shared encoder width
k_{nn} candidates	8	8	Initial neighborhood size
Maximum candidate radius $d_{\max}$	1500 m	not used	WTB proximity fallback / candidate filter
Wake-cone half-angle ϕ	25°	not used	WTB downstream activation
Streamwise decay α	1200 m	not used	WTB wake attenuation
Cross-stream decay β	400 m	not used	WTB wake attenuation
Retained inbound edges M	5	symmetric k_{nn} graph	Final graph sparsity
Optimizer	AdamW	AdamW	Shared optimizer
Learning rate	2×10^{-3}	2×10^{-3}	Shared schedule
Weight decay	10^{-4}	10^{-4}	Shared regularization
Batch size	16	16	Shared mini-batch size
Gradient clipping	1.0	1.0	Shared training stabilization

9 Appendix A. Training and implementation details

9.1 Training loop summary

The optimization procedure is identical across runs except for the routing penalties that are activated in each setting and the anchor labels that are available.

```
Algorithm A1  Physics-aligned MoE training
Input: history windows X, graph sequence A, targets Y, primary regime anchors R,
       auxiliary wake labels W (WTB only), model parameters theta
for each minibatch do
    Build node-level graph context from the precomputed graph tensors
    Encode X with the directed-diffusion GRU to obtain node context H
    Compute gate logits Z and soft routing probabilities G
    Compute expert forecasts and aggregate them with G
    Evaluate L_pred
    If MoE mode is active:
        add L_bal on top-K routing statistics
        add L_align on valid primary regime anchors
        if WTB: add L_force on MPPT/pitch samples
        if WTB: add L_aux on wake labels
        add L_smooth on the final graph snapshot
    Update theta with AdamW and mixed precision
end for
Select the checkpoint with the best validation RMSE
```

9.2 Shared constants

9.3 Dataset-specific thresholds and routing weights

Bai, L., Yao, L., Li, C., Wang, X., Wang, C., 2020. Adaptive graph convolutional recurrent network for traffic forecasting, in: Advances in Neural Information Processing Systems. pp. 17804–17815.

Daenens, S., Verstraeten, T., Daems, P.-J., Nowé, A., Helsen, J., 2025. Spatio-temporal graph neural networks for power prediction in offshore wind farms using SCADA data. Wind Energy Science 10,

Table A2. Dataset-specific regime thresholds used to construct routing anchors.

Dataset	Threshold	Value and meaning
WTB	u_{idle}	3.0 m s^{-1}, idle / active split
WTB	u_{rated}	10.5 m s^{-1}, MPPT / pitch boundary
WTB	p_{th}	2.0 deg, pitch-angle threshold
WTB	$q_{0.75}^{\text{wake}}$	1.0690, upper-quartile wake-score threshold
ERA5	$\varepsilon_{\text{sshf}}$	169,116, 10th percentile of $\lvert\texttt{sshf}\rvert$ on the training split
ERA5	$q_{0.95}^{\Delta}$	458,287.125, 95th percentile of $\lvert\Delta\texttt{sshf}\rvert$ on the training split

Table A3. Active routing-loss weights in the reported corrected models.

Weight	WTB	ERA5	Role
λ_{bal}	1000	0.01	Expert-usage stabilization
λ_{align}	5000	0.2	Coarse regime alignment
λ_{force}	10000	0	Boundary-focused forcing
λ_{aux}	250	0	Wake-sensitive auxiliary routing
λ_{smooth}	0.05	0.05	Local routing coherence

1137–1152. https://doi.org/10.5194/wes-10-1137-2025

Fedus, W., Zoph, B., Shazeer, N., 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. Journal of Machine Learning Research 23, 1–39.

Guo, S., Lin, Y., Feng, N., Song, C., Wan, H., 2019. Attention based spatial-temporal graph convolutional networks for traffic flow forecasting, in: Proceedings of the AAAI Conference on Artificial Intelligence. pp. 922–929. https://doi.org/10.1609/aaai.v33i01.3301922

Haq, I.U., Kumar, A., Rathore, P.S., 2025. Machine learning approaches for wind power forecasting: A comprehensive review. Discover Applied Sciences 7, 1139. https://doi.org/10.1007/s42452-025-07675-x

Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J.-R., Bonavita, M., De Chiara, G., Dahlgren, P., Dee, D., Diamantakis, M., Dragani, R., Flemming, J., Forbes, R., Fuentes, M., Geer, A., Haimberger, L., Healy, S., Hogan, R., Hólm, E., Janisková, M., Keeley, S., Laloyaux, P., Lopez, P., Lupu, C., Radnoti, G., Rosnay, P. de, Rozum, I., Vamborg, F., Villaume, S., Thépaut, J.-N., 2020. The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society 146, 1999–2049. https://doi.org/10.1002/qj.3803

Jacobs, R.A., Jordan, M.I., Nowlan, S.J., Hinton, G.E., 1991. Adaptive mixtures of local experts. Neural Computation 3, 79–87. https://doi.org/10.1162/neco.1991.3.1.79

Jordan, M.I., Jacobs, R.A., 1994. Hierarchical mixtures of experts and the EM algorithm. Neural Computation 6, 181–214. https://doi.org/10.1162/neco.1994.6.2.181

Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang, L., 2021. Physics-informed machine learning. Nature Reviews Physics 3, 422–440. https://doi.org/10.1038/s42254-021-00314-5

Karpatne, A., Atluri, G., Faghmous, J.H., Steinbach, M., Banerjee, A., Ganguly, A., Shekhar, S., Samatova, N.F., Kumar, V., 2017. Theory-guided data science: A new paradigm for scientific discovery from data. IEEE Transactions on Knowledge and Data Engineering 29, 2318–2331. https://doi.org/10.1109/TKDE.2017.2720168

Khodayar, M., Wang, J., 2019. Spatio-temporal graph deep neural network for short-term wind speed

forecasting. IEEE Transactions on Sustainable Energy 10, 670–681. https://doi.org/10.1109/TSTE.2018.2844102

Li, S., Li, X., Jiang, Y., Yang, Q., Lin, M., Peng, L., Yu, J., 2025. A novel frequency-domain physics-informed neural network for accurate prediction of 3D spatio-temporal wind fields in wind turbine applications. Applied Energy 386, 125526. https://doi.org/10.1016/j.apenergy.2025.125526

Li, Y., Yu, R., Shahabi, C., Liu, Y., 2018. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting, in: International Conference on Learning Representations.

Park, J., Park, J., 2019. Physics-induced graph neural network: An application to wind-farm power estimation. Energy 187, 115883. https://doi.org/10.1016/j.energy.2019.115883

Parsa, S.M., 2025. Physics-informed machine learning meets renewable energy systems: A review of advances, challenges, guidelines, and future outlooks. Applied Energy 402, 126925. https://doi.org/10.1016/j.apenergy.2025.126925

Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. Journal of Computational Physics 378, 686–707. https://doi.org/10.1016/j.jcp.2018.10.045

Read, J.S., Jia, X., Willard, J., Appling, A.P., Zwart, J.A., Oliver, S.K., Karpatne, A., Hansen, G.J.A., Hanson, P.C., Watkins, W., Steinbach, M., Kumar, V., 2019. Process-guided deep learning predictions of lake water temperature. Water Resources Research 55, 9173–9190. https://doi.org/10.1029/2019WR024922

Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., Dean, J., 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, in: International Conference on Learning Representations.

Shi, S., Han, L., Liu, Y., others, 2025. Time-MoE: Billion-scale time series foundation models with mixture of experts, in: International Conference on Learning Representations.

Wu, Z., Pan, S., Long, G., Jiang, J., Chang, X., Zhang, C., 2020. Connecting the dots: Multivariate time series forecasting with graph neural networks, in: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 753–763. https://doi.org/10.1145/3394486.3403118

Wu, Z., Pan, S., Long, G., Jiang, J., Zhang, C., 2019. Graph WaveNet for deep spatial-temporal graph modeling, in: Proceedings of the 28th International Joint Conference on Artificial Intelligence. pp. 1907–1913. https://doi.org/10.24963/ijcai.2019/264

Yu, B., Yin, H., Zhu, Z., 2018. Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting, in: Proceedings of the 27th International Joint Conference on Artificial Intelligence. pp. 3634–3640. https://doi.org/10.24963/ijcai.2018/505

Yu, M., Zhang, Z., Li, X., Yu, J., Gao, J., Liu, Z., You, B., Zheng, X., Yu, R., 2020. Superposition graph neural network for offshore wind power prediction. Future Generation Computer Systems 113, 145–157. https://doi.org/10.1016/j.future.2020.06.024

Zehtabiyan-Rezaie, N., Iosifidis, A., Abkar, M., 2023. Physics-guided machine learning for wind-farm power prediction: Toward interpretability and generalizability. PRX Energy 2, 013009. https://doi.org/10.1103/PRXEnergy.2.013009

Zhang, J., Zhao, X., 2021. Spatiotemporal wind field prediction based on physics-informed deep learning and LIDAR measurements. Applied Energy 288, 116641. https://doi.org/10.1016/j.apenergy.2021.116641

Zhou, J., Lu, X., Xiao, Y., Tang, J., Su, J., Li, Y., Liu, J., Lyu, J., Ma, Y., Dou, D., 2024. SDWPF: A dataset for spatial dynamic wind power forecasting over a large turbine array. Scientific Data 11, 649. https://doi.org/10.1038/s41597-024-03427-5